

Lab Exercise no. 1.0: Basic LAN Connectivity with 25 Hosts

Objective:

- Design a simple Local Area Network (LAN) with 25 host computers connected to a switch.
- Assign unique IP addresses to each host.
- Verify connectivity using the ping command.
- Document device information (device name, MAC address, port, IP address).

Materials:

Cisco Packet Tracer software.

Procedure:

Part 1: Network Topology Design

- Open Cisco Packet Tracer: Launch the Cisco Packet Tracer application.
- Add a Switch:
 - From the device panel at the bottom, select "Network Devices."
 - Choose "Switches."
 - Select a 24-port switch (e.g., 2960) and drag it onto the workspace.
 - Since you need to connect 25 computers you will need to add a second switch and connect them via a fastethernet port, or use a 48 port switch.
- Add Host Computers:
 - From the device panel, select "End Devices."
 - Choose "PC" and drag 25 PCs onto the workspace.
 - Arrange the PCs around the switch(es) for clarity.
- Connect the Devices:
 - Select "Connections" (the lightning bolt icon).
 - Choose "Copper Straight-Through" cable.
 - Connect each PC to an available FastEthernet port on the switch(es). Start with PC0 to FastEthernet0/1, PC1 to FastEthernet0/2, and so on.
 - If using two switches, connect one of the fastethernet ports from each switch

together. Part 2: IP Addressing

- IP Addressing Scheme:
 - Use the private IP address range 192.168.1.0/24.
 - Assign IP addresses sequentially, starting with 192.168.1.1 for PC0, 192.168.1.2 for PC1, and so on, up to 192.168.1.25 for PC24.
 - The subnet mask will be 255.255.255.0.
- Configure IP Addresses:
 - Double-click on PC0.
 - Go to the "Desktop" tab.
 - Select "IP Configuration."

- Enter the IP address (192.168.1.1), subnet mask (255.255.255.0), and leave the default gateway blank (for this simple LAN).
- Close the IP Configuration window.
- Repeat this process for all 25 PCs, incrementing the IP address for

each. Part 3: Verification

- Ping Test:
- Double-click on PC0.
- Go to the "Desktop" tab.
- Select "Command Prompt."
- Type ipconfig and press Enter to verify the IP address assigned to PC0.
- Type ping 192.168.1.2 (the IP address of PC1) and press Enter.
- If the ping is successful, you will see replies from 192.168.1.2.
- Repeat the ping test from PC0 to several other PCs.
- Repeat the ping test from other PC's to other PC's in the network.
- MAC Address and Port Information:
- Double click on each PC.
- Go to the command prompt, and type ipconfig /all. This will display the MAC address.
- By hovering your mouse over the cable connection on the PC and the switch, you can see the port numbers utilized.
- Record the device name, MAC address, port number, and IP address for each PC in a

table. Part 4: Documentation

- Create a Table:
- Create a table in a document or spreadsheet with the following columns:
- Device Name (e.g., PC0, PC1, etc.)
- MAC Address
- Switch Port (e.g., FastEthernet0/1, FastEthernet0/25 etc.)
- IP Address
- Fill in the table with the information gathered during the verification step.
- Packet Tracer File:
- Save the Packet Tracer file with a descriptive name (e.g., "Basic_LAN_25_Hosts.pkt").

Example Table (Partial):

Device Name	MAC Address	Switch Port	IP Address
Server1	00:0A:95:9D:68:16	FastEthernet0/1	192.168.1.1
Server2	00:0A:95:9D:68:17	FastEthernet0/2	192.168.1.2
PC1	00:0A:95:9D:68:18	FastEthernet0/3	192.168.1.3
PC24	00:0A:95:9D:68:3A	FastEthernet0/25	192.168.1.25

Export to Sheets

Lab Report:

Students should submit:

1. The Packet Tracer file (.pkt).
2. The completed documentation table.
3. A brief description of the learning takeaways and observations made.
4. Screenshots of successful ping tests. (2 or 3)